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Branding is what people say about you when you are not in the room.



-Jeff Bezos



Fig. 6: Suggested enclosure

connected across the variable output to continuously display the output voltage. The adjustable regulated output is available at connector CON4.

Construction and testing

An actual-sized, single-sided PCB layout for the universal power supply is shown in Fig. 4, and its component layout is shown in Fig. 5. After assembling the circuit on the PCB, mount it inside a suitable enclosure, as shown in Fig. 6, for reliable operation and protection.

Alternatively, the circuit can be assembled on a general-purpose PCB or a custom-designed PCB. While designing the PCB, tracks carrying higher current should

be made wider to reduce voltage drop and heating. Proper heat sinks must be used for the LM7812, LM7912, and especially the LM338 because these regulators dissipate significant heat during operation. The transformer should be mounted securely

inside a suitable enclosure, and insulated wires must be used for all AC mains connections. Care should also be taken to separate high-voltage AC wiring from low-voltage DC wiring. Ensure correct polarity while connecting electrolytic capacitors and use a good-quality preset for stable voltage adjustment.

After assembling the circuit, carefully inspect all solder joints and component orientations before applying power. Initially, do not connect any load to the outputs. The circuit should initially be powered through a series bulb tester for additional safety during testing.

Verify the regulated outputs at CON2 and CON3 to confirm proper +12V and -12V operation. Adjust preset VR1 and observe the output voltage on the voltmeter. The voltage should vary smoothly from approximately 1.2V to around 22V. If excessive heating is observed during operation, larger heat sinks or a cooling fan may be used.

Caution. Never touch the AC section while the circuit is powered. A fuse should be connected to the transformer primary for protection against short circuits and overload conditions. Ensure adequate ventilation for the voltage regulators. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-universal-power-supply>

PARTS LIST

Semiconductors:

- IC1 - LM7812 positive voltage regulator
- IC2 - LM7912 negative voltage regulator
- IC3 - LM338 adjustable voltage regulator
- BR1 - 1A bridge rectifier
- D1 - 1N4007 rectifier diode
- DS1 - DC digital voltmeter module

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 470 Ω
- R2 - 150 Ω
- R3 - 120 Ω
- VR1 - 5k Ω potentiometer

Capacitors:

- C1, C2 - 1000 μ F, 25V electrolytic
- C3-C6 - 2.2 μ F electrolytic

Miscellaneous:

- X1 - 230V AC primary to 15V-0-15V, 1A step-down transformer
- CON1-CON4 - 2-pin screw terminal connectors
- PCB - Single-side PCB
- Heat sinks for IC1, IC2, and IC3
- Connecting wires, enclosure, and solder

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